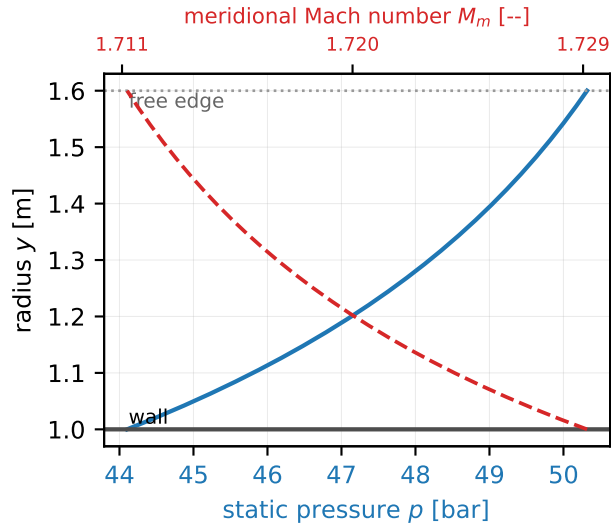
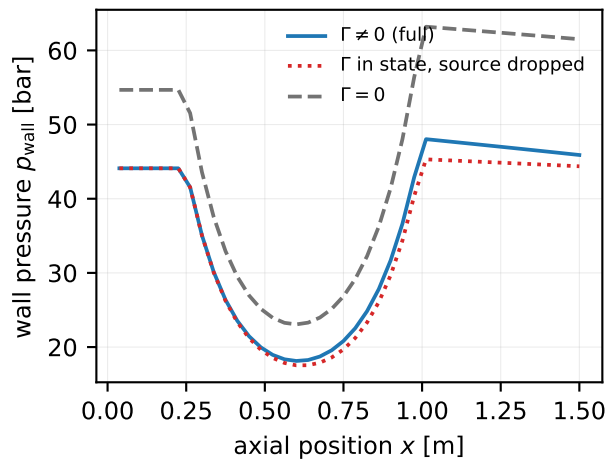


(a) the equilibrium duct: uniform u ,
all the structure in the swirl energy term



(b) the expansion world: the two alive
controls --- swirl and its source both matter



(c) what the free vortex is: swirl falling
as $1/y$, one constant Γ for the flow

